HW#1 solutions

1. T = 1/f. So, T = 1/(0.5x10^9) = 2GHz. If the clock cycle time gets reduced by 20%, it becomes 0.5ns – 0.5ns x 20% = 0.4ns. So, the resulting clock frequency becomes 1/0.4ns = 2.5GHz.

2. CPU time = Instruction count x average CPI x Clock cycle time. So, 10^6 x 2.5 x 0.5ns = 1.25ms.

3. Overclocked clock cycle time = 1/2.5GHz = 0.4ns.

CPU time = 10^6 x 2.5 x 0.4ns = 1ms. Performance ratio = 1.25/1 = 1.25. So, it is 1.25 times faster.

4. Performance ratio = 2 = (Instruction count x 3.0 x 0.5ns) / (Instruction count x 2.0 x CPU A clock cycle time). So, we can cancel Instruction count and solve this equation to find CPU A clock cycle time = 0.375ns. Therefore, CPU A clock frequency = 1/0.375ns = 2.666GHz.

5. Since the execution time of the program on the original machine is not specified, let us represent it as $T_{orginal}$.

Improvement factor = 10.
$T_{affected} = 0.5 \times T_{original}$ and $T_{unaffected} = 0.5 \times T_{original}$.
$T_{improved} = (0.5 \times T_{original})/10 + 0.5 \times T_{original}$
Performance ratio = $T_{original} / T_{improved} = 1.8181$.
So, the improved machine is 1.8181 times faster than the original machine.

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